

Deep Generative Models (25120)

Problem Set 5

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(*) starred problems are optional!

1 Ergodicity Conditions in MCMC

In this part we study general Markov chains and how they can be used for MCMC sampling.

Let $(X_n)_{n \geq 0}$ be a discrete-time Markov chain with state space $\mathcal{X}$ (finite or countable) and transition kernel

$$P(x, y) = \mathbb{P}(X_{n+1} = y \mid X_n = x), \quad x, y \in \mathcal{X}.$$

A probability distribution π on $\mathcal{X}$ is called *stationary* for P if

$$\pi P = \pi, \quad \text{i.e.} \quad \sum_{x \in \mathcal{X}} \pi(x) P(x, y) = \pi(y) \quad \text{for all } y \in \mathcal{X}.$$

A probability distribution π is called a *limiting distribution* of the chain if, for every initial distribution μ_0 on $\mathcal{X}$,

$$\mu_0 P^n \xrightarrow{n \rightarrow \infty} \pi \quad (\text{convergence of distributions}).$$

A standard ergodic theorem for discrete-time Markov chains on countable state spaces says that if the chain is **irreducible**, **aperiodic**, and **positive recurrent**, then:

1. there exists a *unique* stationary distribution π , and
2. for any initial state x , the distribution of X_n converges to π as $n \rightarrow \infty$.

You may take this theorem as given (no proof required).

Part (a): Conditions and Their Impact on MCMC

We recall the three conditions.

Irreducible. For states $x, y \in \mathcal{X}$, say that x *can reach* y if there exists an integer $n \geq 0$ such that

$$P^n(x, y) > 0.$$

We say x and y *communicate* if each can reach the other.

The Markov chain is called *irreducible* if all states communicate with each other:

$$\forall x, y \in \mathcal{X}, \quad x \text{ can reach } y \text{ and } y \text{ can reach } x.$$

Aperiodic. For a state $x \in \mathcal{X}$, define its *period* by

$$d(x) = \gcd\{n \geq 1 : P^n(x, x) > 0\}.$$

We say that x is *aperiodic* if $d(x) = 1$. If the chain is irreducible, all states share the same period; in that case, we say the chain is *aperiodic* if one (and hence every) state has period 1.

Positive recurrent. For a state $x \in \mathcal{X}$, define the *first return time*

$$T_x = \inf\{n \geq 1 : X_n = x \mid X_0 = x\}.$$

The state x is called *recurrent* if $\mathbb{P}_x(T_x < \infty) = 1$. A recurrent state x is called *positive recurrent* if

$$\mathbb{E}_x[T_x] < \infty.$$

In an irreducible chain, either all states are positive recurrent or none are. We say the chain is *positive recurrent* if (in the irreducible case) every state is positive recurrent.

Answer the following.

(1) **Impact of each condition on MCMC sampling.**

For each of the three conditions above (irreducible, aperiodic, positive recurrent), briefly explain:

- why this condition is important for using the chain to sample from a target distribution
- what can go wrong for sampling if this condition fails.

You may use simple toy examples (e.g. a two-state chain that oscillates, or a random walk on $\mathbb{Z}$) to illustrate your answers.

(2) **Why all three together?**

Using the ergodic theorem stated above, explain why having *all three* conditions (irreducible, aperiodic, positive recurrent) is exactly what we want for MCMC:

- in terms of the existence and uniqueness of a stationary distribution
- in terms of convergence of the chain to that distribution from any starting point.

Part (b): Detailed Balance and the MH Kernel

In many MCMC algorithms we design a Markov kernel P to satisfy the *detailed balance* condition with respect to a given target distribution π :

$$\pi(x) P(x, y) = \pi(y) P(y, x) \quad \text{for all } x, y.$$

- (1) What does the detailed balance condition guarantee about the stationary distribution of the Markov chain with transition kernel P and target π ?
- (2) In the Metropolis–Hastings algorithm, we first choose a proposal kernel $q(x, y)$ and then define an MH transition kernel $P_{\text{MH}}(x, y)$ using the usual acceptance probability

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right).$$

What does this construction guarantee about the stationary distribution of the MH chain?

2 Score-Matching Variants

In this Question we study *objective equivalence* in score-based learning: an *explicit* objective that depends on the unknown data score (or its noisy variant) can be rewritten as an *implicit* objective that does not. We will prove this equivalence in three settings: (1) Hyvärinen's score matching (L_{ESM} vs. L_{ISM}), (2) Vincent's denoising score matching (L_{ESM,q_σ} vs. L_{DSM,q_σ}), and (3) Song's sliced score matching (L_{ESSM} vs. L_{ISSM}), including an identifiability result and special choices of p_v where sliced and full objectives coincide.

Part 1: Explicit vs. Implicit Score Matching

Let $x \in \mathbb{R}^d$ have data density $p_{\text{data}}(x)$. Define:

$$s_\theta(x) \triangleq \nabla_x \log p_\theta(x), \quad s_{\text{data}}(x) \triangleq \nabla_x \log p_{\text{data}}(x).$$

$$L_{\text{ESM}}(\theta) \triangleq \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} [\|s_\theta(x) - s_{\text{data}}(x)\|_2^2], \quad L_{\text{ISM}}(\theta) \triangleq \mathbb{E}_{x \sim p_{\text{data}}} \left[\frac{1}{2} \|s_\theta(x)\|_2^2 + \text{tr}(\nabla_x s_\theta(x)) \right].$$

Assume: $p_{\text{data}}(x)$ is differentiable, $\mathbb{E}_{x \sim p_{\text{data}}} [\|s_\theta(x)\|_2^2]$ and $\mathbb{E}_{x \sim p_{\text{data}}} [\|s_{\text{data}}(x)\|_2^2]$ are finite for any θ , and $p_{\text{data}}(x) s_\theta(x) \rightarrow 0$ for any θ as $\|x\|_p \rightarrow \infty$.

(a) Fix an index $i \in \{1, \dots, d\}$. Treat x_{-i} as fixed and prove the identity

$$\lim_{a \rightarrow \infty, b \rightarrow -\infty} \left(f(x_{-i}, a) g(x_{-i}, a) - f(x_{-i}, b) g(x_{-i}, b) \right) = \int_{-\infty}^{\infty} f(x) \frac{\partial g(x)}{\partial x_i} dx_i + \int_{-\infty}^{\infty} g(x) \frac{\partial f(x)}{\partial x_i} dx_i.$$

(b) Using part (a) and the assumptions, prove that $L_{\text{ESM}}(\theta) = L_{\text{ISM}}(\theta) + C$ for a constant C independent of θ .

Part 2: Noisy ESM vs. DSM

Let $q_\sigma(\tilde{x} | x)$ be a differentiable kernel, and define

$$q_\sigma(\tilde{x}) \triangleq \int p_{\text{data}}(x) q_\sigma(\tilde{x} | x) dx, \quad q_\sigma(x, \tilde{x}) \triangleq p_{\text{data}}(x) q_\sigma(\tilde{x} | x).$$

$$L_{\text{ESM},q_\sigma}(\theta) \triangleq \mathbb{E}_{\tilde{x} \sim q_\sigma} \left[\frac{1}{2} \|s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log q_\sigma(\tilde{x})\|_2^2 \right],$$

$$L_{\text{DSM},q_\sigma}(\theta) \triangleq \mathbb{E}_{(x,\tilde{x}) \sim q_\sigma(x,\tilde{x})} \left[\frac{1}{2} \|s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log q_\sigma(\tilde{x} | x)\|_2^2 \right].$$

(a) Prove that

$$S(\theta) \triangleq \mathbb{E}_{\tilde{x} \sim q_\sigma} [\langle s_\theta(\tilde{x}), \nabla_{\tilde{x}} \log q_\sigma(\tilde{x}) \rangle] = \mathbb{E}_{(x,\tilde{x}) \sim q_\sigma(x,\tilde{x})} [\langle s_\theta(\tilde{x}), \nabla_{\tilde{x}} \log q_\sigma(\tilde{x} | x) \rangle].$$

(b) Using part (a), show that $L_{\text{ESM},q_\sigma}(\theta) = L_{\text{DSM},q_\sigma}(\theta) + \text{const}$ (constant independent of θ).

* Part 3: Sliced Score Matching

Let $v \in \mathbb{R}^D$ be drawn from p_v . Define the sliced explicit and implicit objectives

$$L_{\text{ESSM}}(\theta) \triangleq \frac{1}{2} \mathbb{E}_{v \sim p_v} \mathbb{E}_{x \sim p_{\text{data}}} \left[(v^\top s_\theta(x) - v^\top s_{\text{data}}(x))^2 \right],$$

$$L_{\text{ISSM}}(\theta) \triangleq \mathbb{E}_{v \sim p_v} \mathbb{E}_{x \sim p_{\text{data}}} \left[\frac{1}{2} (v^\top s_\theta(x))^2 + v^\top (\nabla_x s_\theta(x)) v \right]$$

Assume: $p_{\text{data}}(x)$ and $p_\theta(x)$ are positive and differentiable; the expectations in this problem are finite; $\mathbb{E}_{v \sim p_v} \|v\|_2^2 < \infty$; and $s_\theta(x) p_{\text{data}}(x) \rightarrow 0$ as $\|x\| \rightarrow \infty$. For part (c), also assume the model is well-specified and identifiable: $\exists \theta^*$ with $p_{\theta^*} = p_{\text{data}}$ and $p_{\theta_1} = p_{\theta_2} \Rightarrow \theta_1 = \theta_2$, and assume $\Sigma_v \triangleq \mathbb{E}_{p_v} [vv^\top] \succ 0$.

(a) Prove the identity

$$-\mathbb{E}_{v \sim p_v} \mathbb{E}_{x \sim p_{data}} \left[(v^\top s_\theta(x)) (v^\top s_{data}(x)) \right] = \mathbb{E}_{v \sim p_v} \mathbb{E}_{x \sim p_{data}} \left[v^\top (\nabla_x s_\theta(x)) v \right].$$

(b) Using part (a), prove that $L_{\text{ESSM}}(\theta) = L_{\text{ISSM}}(\theta) + C$ for a constant C independent of θ .

(c) Under the assumptions stated above for part (c), prove that

$$L(\theta; p_v) \triangleq \frac{1}{2} \mathbb{E}_{v \sim p_v} \mathbb{E}_{x \sim p_{data}} \left(v^\top (s_\theta(x) - s_{data}(x)) \right)^2 = 0 \iff \theta = \theta^*.$$

(d) Let $\Delta s(x) \triangleq s_\theta(x) - s_{data}(x)$ and

$$L_{\text{ESSM}}(\theta) = \frac{1}{2} \mathbb{E}_{v \sim p_v} \mathbb{E}_{x \sim p_{data}} \left[(v^\top \Delta s(x))^2 \right], \quad L_{\text{ESM}}(\theta) = \frac{1}{2} \mathbb{E}_{x \sim p_{data}} [\|\Delta s(x)\|_2^2].$$

(i) Show that if $\mathbb{E}_{p_v}[vv^\top] = I_D$, then $L_{\text{ESSM}}(\theta) = L_{\text{ESM}}(\theta)$.

(ii) Verify $\mathbb{E}[vv^\top] = I_D$ when v_i are i.i.d. Rademacher with $\mathbb{P}(v_i = \pm 1) = \frac{1}{2}$ or $v \sim \mathcal{N}(0, I_D)$.

3 Score-based Models

1. We know that, given an estimate of the score function (the gradient of the log probability density), we can use the Langevin dynamics algorithm to generate samples from the desired probability distribution. Explain in full detail why, in practice, this method encounters problems. What methods are used to solve these problems?
2. Suppose we want to build a conditional model for generating samples from different classes. Explain how, given a pretrained *unconditional* score-based model together with a soft classifier, we can do this. Write down the corresponding equations.
3. In constructing the conditional model using the method in part (2), why can we not use a frozen pretrained classifier? In other words, explain why the classifier must indeed be trained.
4. The training objective for score-based models, after some simplifications, becomes

$$J(\theta) = \mathbb{E}_{p_{data}} \left[\frac{1}{2} \|s_\theta(x)\|_2^2 + \text{tr}(\nabla_x s_\theta(x)) \right],$$

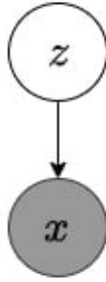
where s_θ is our estimate of the score function. Explain in full detail the intuition behind each term in the above expression.

4 Energy-based Models

1. Explain the Contrastive Divergence method for training energy-based models. Why is this method very time-consuming?
2. State one advantage and one disadvantage of Rejection Sampling compared to the Metropolis-Hastings method.
3. Consider the following mixture of Gaussians distribution:

$$p(x) = 0.6 \mathcal{N}(50, 1) + 0.4 \mathcal{N}(0.001).$$

If the proposal distribution is $Q(x'|x) = \mathcal{N}(x, 0.5)$, explain why the MH algorithm is extremely slow for generating samples from the above distribution (state two reasons). What solutions do you propose to address this issue?



5 Score Matching

Consider the Bayesian network together with the following conditional distributions:

$$p(z) = e^{-z}, \quad z \geq 0, \quad p(x | z) = z x e^{-\frac{zx^2}{2}}, \quad x \geq 0.$$

We want to approximate the posterior distribution $p(z | x)$ with

$$q(z | x) = \theta^2 z e^{-\theta z}, \quad z \geq 0.$$

Determine the optimal value of the parameter θ using score matching.